

ENGINEERING CALCULATIONS

Adaptive Power Distribution System

12 V DC Input • Regulated 5 V & 3.3 V Outputs

1. Purpose and Design Basis

Purpose. This document presents the calculations used to size the loads, determine output power and current, assess regulator requirements, justify protection, and consider power loss and thermal behavior.

Assessment basis. The assessment specifies a 12 V DC input, regulated 5 V and 3.3 V outputs, Load A = 5 V/500 mA, Load B = 3.3 V/300 mA, and Load C = 5 V/1 A. It also asks for calculations covering current, power, losses, regulator requirements and thermal considerations.

Implemented design. The completed design uses LM2596S-5.0 and LM2596S-3.3 buck-converter stages. The Multisim loads are represented by 10 Ω , 11 Ω and 5 Ω resistors.

2. Design Input and Load Requirements

Load/Rail	Voltage	Current	Equivalent R	Power
Load A	5.0 V	0.50 A	10 Ω	2.50 W
Load B	3.3 V	0.30 A	11 Ω	0.99 W
Load C	5.0 V	1.00 A	5 Ω	5.00 W
5 V total	5.0 V	1.50 A	—	7.50 W
3.3 V total	3.3 V	0.30 A	—	0.99 W
Total output	—	—	—	8.49 W

3. Load Resistance Calculations

3.1 Load A — 5 V, 500 mA

$$R = V / I = 5 / 0.5 = 10 \Omega$$

$$P = V \times I = 5 \times 0.5 = 2.50 \text{ W}$$

The simulation therefore uses 10 Ω . The resistor should have a practical power rating above 2.50 W.

3.2 Load B — 3.3 V, 300 mA

$$R = 3.3 / 0.3 = 11 \Omega$$

$$P = 3.3 \times 0.3 = 0.99 \text{ W}$$

The simulation uses 11 Ω . A 1 W resistor is the minimum nominal rating; a higher rating provides additional margin.

3.3 Load C — 5 V, 1 A

$$R = 5 / 1 = 5 \Omega$$

$$P = 5 \times 1 = 5.00 \text{ W}$$

The simulation uses 5 Ω . The resistor should be rated above 5 W for practical operation.

4. Output Current and Power Requirements

4.1 5 V rail

$$I_{5V,\text{total}} = 0.50 + 1.00 = 1.50 \text{ A}$$

$$P_{5V} = 5 \times 1.50 = 7.50 \text{ W}$$

4.2 3.3 V rail

$$I_{3.3V,\text{total}} = 0.30 \text{ A}$$

$$P_{3.3V} = 3.3 \times 0.30 = 0.99 \text{ W}$$

4.3 Total regulated output

$$P_{\text{OUT},\text{total}} = 7.50 + 0.99 = 8.49 \text{ W}$$

Therefore, the system is designed to deliver approximately 8.49 W when all specified loads operate simultaneously at nominal voltage and current.

5. Input Power and Current

The assessment does not specify a guaranteed converter efficiency. Therefore, exact input current and loss should be taken from the final simulation or hardware measurement. The governing relationships are:

$$P_{\text{IN}} = P_{\text{OUT}} / \eta$$

$$I_{\text{IN}} = P_{\text{IN}} / V_{\text{IN}}$$

Assumed η	Input power	Input current @ 12 V	Estimated loss
80%	10.61 W	0.884 A	2.12 W
85%	9.99 W	0.833 A	1.50 W
90%	9.43 W	0.786 A	0.94 W

Reporting note: The efficiency cases above are engineering scenarios, not measured values. Replace them with actual Multisim input measurements if available.

6. Regulator Requirement

Regulator	Nominal output	Required load current	Topology	Requirement
LM2596S-5.0	5.0 V	1.50 A	Buck	Must support ≥ 1.50 A
LM2596S-3.3	3.3 V	0.30 A	Buck	Must support ≥ 0.30 A

A buck converter is appropriate because both required output voltages are lower than the 12 V input. Compared with a linear regulator, a switching buck topology substantially reduces conversion dissipation at these load currents.

7. Protection Calculations

7.1 Input fuse

The design uses a 3 A input fuse. Using the 80% illustrative efficiency case:

$$I_{IN} \approx 10.61 / 12 = 0.884 \text{ A}$$

$$3 \text{ A} / 0.884 \text{ A} \approx 3.39$$

Thus the selected fuse is above the estimated normal full-load input current while providing a protective limit for abnormal input overcurrent. Final fuse selection should also consider startup/inrush current and the selected fuse's time-current characteristic.

7.2 Output protection

Branch	Specified load	Protection	Purpose
5 V / Load A	0.50 A	0.5 A polyfuse	Branch overcurrent protection
5 V / Load C	1.00 A	1.0 A polyfuse	Branch overcurrent protection
3.3 V / Load B	0.30 A	Low-current polyfuse	Branch overcurrent protection

Polyfuse hold/trip behavior depends on the exact part, ambient temperature and time; the final BOM should therefore reference the selected manufacturer's datasheet.

8. Passive Components and Energy Storage

Input bulk capacitor. 470 μ F is used on the protected 12 V rail for local energy storage and input-voltage smoothing.

High-frequency bypass. 100 nF capacitors provide local high-frequency bypassing around the converter stages.

Output capacitors. 330 μ F capacitors are used on the 5 V and 3.3 V rails to support transient load demand and reduce ripple.

Inductors. 68 μ H inductors are used in both buck stages. The exact final component should be verified for saturation current, RMS current and voltage rating against the chosen LM2596 implementation.

9. Thermal Considerations

The specified load consumes 8.49 W at the regulated outputs. Converter heat is determined by actual input power:

$$P_{\text{LOSS}} = P_{\text{IN}} - P_{\text{OUT}}$$

For reference, the combined converter loss would be about 2.12 W at 80% efficiency, 1.50 W at 85%, or 0.94 W at 90%. These are scenarios rather than measured results.

The PCB should provide adequate copper around the regulator power and ground connections, keep high-current paths short and wide, and provide suitable clearance and airflow around the regulators and power resistors. The 5 Ω simulation load dissipates 5 W and therefore requires a suitably rated power resistor.

10. Voltage Regulation Verification

The Multisim simulation is used to verify the intended 5.0 V and 3.3 V steady-state outputs. Insert the final steady-state probe values from the simulation in the table below.

Rail	Nominal	Final simulated value	Error
5 V	5.0 V	Insert Multisim value	(Measured–5)/5 $\times$ 100%
3.3 V	3.3 V	Insert Multisim value	(Measured–3.3)/3.3 $\times$ 100%

The earlier Multisim oscilloscope check showed the output reaching the regulated region on a much shorter simulation timescale than the long real-time observation. Therefore, the long timer observation should not be used as the circuit's actual electrical settling-time specification.

11. Summary of Key Calculations

Parameter	Calculated value
5 V total current	1.50 A
3.3 V total current	0.30 A
5 V output power	7.50 W
3.3 V output power	0.99 W
Total output power	8.49 W
Load A	10 Ω , 2.50 W
Load B	11 Ω , 0.99 W
Load C	5 Ω , 5.00 W
Input fuse	3 A